DO NOW — RECALL

Signed binary and shifts

Name: _____ Class: _____ Date: _____

1. Does changing a by-value int parameter change the caller's int?

2. What is a return value?

3. Convert 00001010 from unsigned binary to decimal.

4. Calculate 00001111 + 00000001 using 8 bits.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

No. It is a copy.

2. Model answer

A result a function passes to its caller.

3. Model answer

10.

4. Model answer

00010000, decimal 16.

Vocabulary — know these terms

two's complement — Fixed-width signed binary representation.

sign bit — Leftmost bit indicating sign in two's complement.

logical shift — Shift with zero bits entering vacated positions.

arithmetic right shift — Shift right while extending the sign bit.

overflow — Result cannot be represented in the available bit width.

NEW LEARNING

Read, annotate and explain

POSITIVE 13

+13 written with 8 bits is 00001101.

INVERT BITS

Invert each bit to make 11110010.

ADD ONE

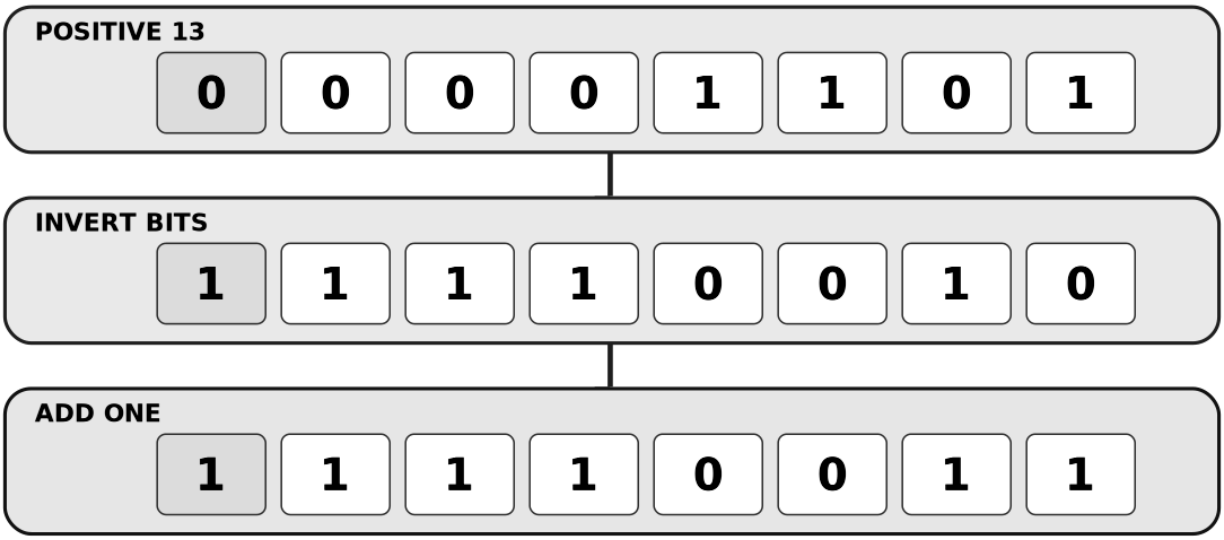
Add one to get 11110011, the 8-bit two's-complement representation of −13.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Predict what happens



POSITIVE 13: 00001101; INVERT BITS: 11110010; ADD ONE: 11110011

I DO — WORKED EXAMPLE

Follow the worked method

Find −9 in 8-bit two's complement.

Predict first

Write +9 in 8 bits and invert all the bits.

Model the result

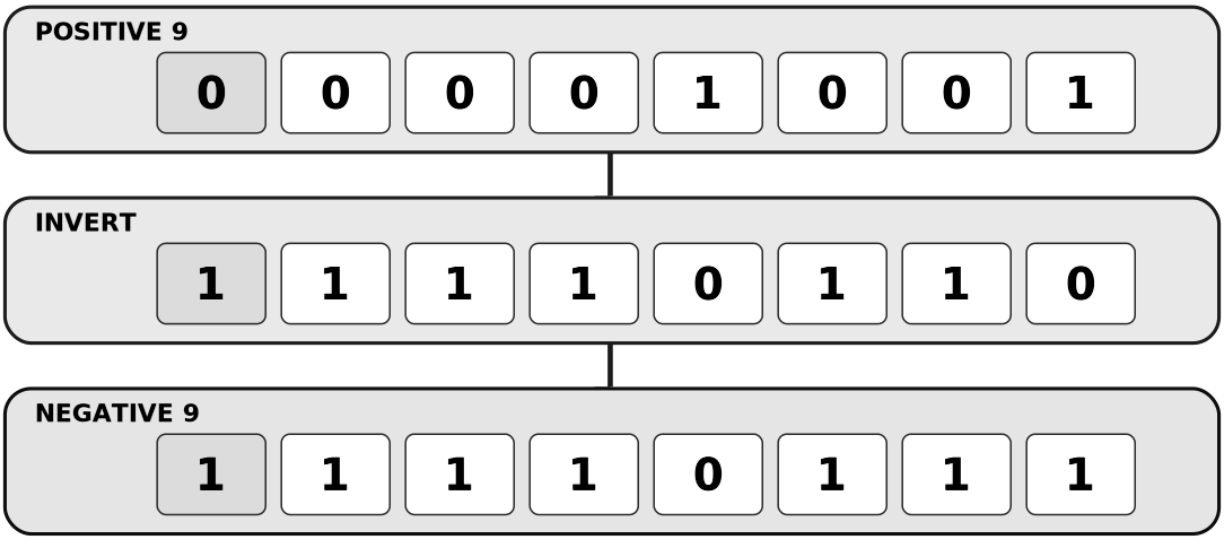
Add 1 to the inverted bits: 11110111 represents −9.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find −9 in 8-bit two's complement.

Write +9 in 8 bits and invert all the bits.



POSITIVE 9: 00001001; INVERT: 11110110; NEGATIVE 9: 11110111

WE DO — GUIDED PRACTICE

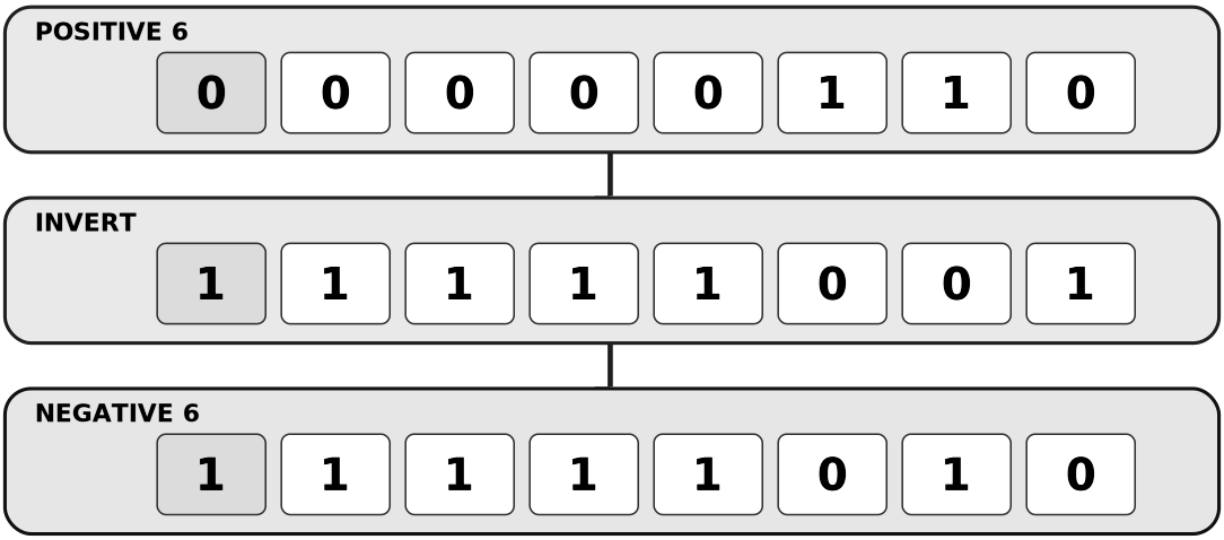
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Convert −6 into 8-bit two's complement.

+6 = 00000110. Invert to get 11111001.



POSITIVE 6: 00000110; INVERT: 11111001; NEGATIVE 6: 11111010

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first +6 = 00000110. Invert to get 11111001.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Write −5 as 8-bit two's complement.

Question 2

Unsigned 8-bit 00110100 is shifted logically right one place. Give the result and decimal value.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

11110100 represents −12 in two's complement. Arithmetically shift right one place and explain the new left bit.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Arithmetic right shift of a negative two's-complement value inserts zero at the left.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model the result

Add 1: 11111010 is the representation of −6.

IMPROVE: Explain one correction to your own reasoning.